

Quiz 3

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Given a system of equations

$$x + 2y + 3z = 1$$

$$2x + 4y - z = 4$$

$$3y + z = 3$$

(a) Write down the augmented matrix $[A, b]$ and which one is the first pivot?

$$[A, b] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & -1 & 4 \\ 0 & 3 & 1 & 3 \end{array} \right]$$

The first pivot is 1.

(b) Find the *Elimination matrix* E_{21} which can be used to eliminate the the first component below the first pivot and calculate $E_{21}[A, b]$ (i.e., $[E_{21}A, E_{21}b]$).

Multipier $\ell_{21} = \frac{2}{1} = 2$, thus $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$E_{21}[A, b] = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 1 \\ 2 & 4 & -1 & 4 \\ 0 & 3 & 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 0 & -7 & 2 \\ 0 & 3 & 1 & 3 \end{bmatrix}$$

(c) Find the *Permutation matrix* P_{23} such that $P_{23}E_{21}[A, b]$ becomes upper triangular.

$$P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$\left(P_{23}E_{21}[A, b] = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & 3 & 1 & 3 \\ 0 & 0 & -7 & 2 \end{bmatrix} \right)$$

Problem 2. (4 points) Let

$$A = \begin{bmatrix} 2 & -4 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 3 \\ 1 & 1 & -2 \end{bmatrix}$$

(a) Compute AB . Is BA possible and why?

$$AB = \begin{bmatrix} 2 & -4 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 & 3 \\ 1 & 1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -4 & 14 \\ 5 & 3 & -3 \end{bmatrix}$$

BA is impossible since B has 3 columns but A only has 2 rows.

(b) Find A^{-1} .

$$A^{-1} = \frac{1}{2 \cdot 3 - (-4) \cdot 1} \begin{bmatrix} 3 & 4 \\ -1 & 2 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 3 & 4 \\ -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{10} & \frac{2}{5} \\ -\frac{1}{10} & \frac{1}{5} \end{bmatrix}$$